

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

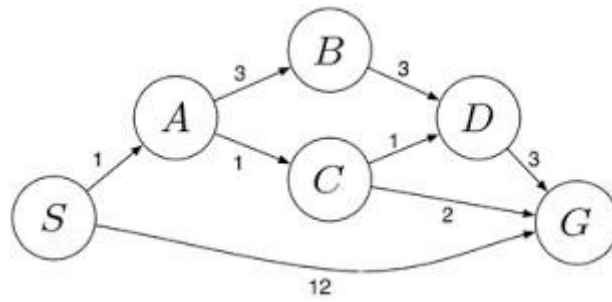
Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

		requirements accurately			
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1.water jug problem

Initial State

- **4-gallon jug (A)** = 0 gallons
- **3-gallon jug (B)** = 0 gallons

Goal: Get exactly **2 gallons** in the **4-gallon jug**.

Step	Action	4-Gallon Jug (A)	3-Gallon Jug (B)
Start	Both jugs empty	0	0
1	Fill the 3-gallon jug	0	3
2	Pour 3-gallon jug into 4-gallon jug	3	0
3	Fill the 3-gallon jug again	3	3
4	Pour from 3-gallon jug into 4-gallon jug until full (only 1gallon fits)	4	2
5	Empty the 4-gallon jug	0	2
6	Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug	2	0

Final State

- **4-gallon jug** = 2 gallons
- **3-gallon jug** = 0 gallons

Hence, the **4-gallon jug** contains exactly **2 gallons** of water.

2.Mars Rover Agent

i. What are the percepts for this agent?

Percepts are the information the Mars Rover receives from its sensors.

- Camera images (terrain, rocks, obstacles)
- Temperature readings
- Soil and rock composition
- Atmospheric conditions (pressure, wind, dust)
- Distance from obstacles (LIDAR/proximity sensors)
- Position and orientation (GPS-like localization, IMU)
- Battery/power level
- Wheel condition and rover health

ii. Characterize the operating environment

The Mars Rover operates in the following environment:

- Partially Observable: Cannot see the entire environment at once.
- Deterministic: Most actions have predictable outcomes, but slippage or dust storms can introduce uncertainty.
- Sequential: Current actions affect future actions.
- Dynamic: Environment changes due to dust storms, lighting, and terrain conditions.
- Continuous: Movement, time, and sensor values are continuous.
- Single-Agent: The rover performs its mission independently.

iii. What are the actions the agent can take?

The Mars Rover can:

- Move forward, backward, left, and right
- Stop or avoid obstacles
- Capture images and videos
- Collect soil and rock samples
- Analyze samples using onboard instruments
- Drill into rocks
- Transmit data and images to Earth
- Monitor battery and recharge using solar panels (if available)
- Perform self-diagnostics

iv. How can one evaluate the performance of the agent?

The rover's performance can be evaluated by:

- Successfully reaching target locations
- Accuracy of scientific observations
- Number and quality of samples collected
- Avoiding obstacles and hazards
- Efficient use of battery power
- Reliable communication with Earth
- Completing mission objectives within the planned time
- Maintaining rover health and safety

v. What sort of agent architecture is most suitable for this agent? Why?

Most Suitable Architecture: Utility-Based Learning Agent

Reason:

- It selects the best action based on the current situation.
- It learns from previous experiences to improve navigation and decision-making.
- It can adapt to unexpected conditions such as rough terrain or dust storms.
- It balances multiple objectives, such as conserving power, avoiding risks, and maximizing scientific discoveries.
- It can make autonomous decisions because communication with Earth is delayed.

3.8-Queens Problem – Problem Formulation

1. Initial State

- The chessboard is empty (no queens placed).

2. State Space

- A state represents the arrangement of queens on the board.
- Each new state is formed by placing one queen in a valid position.
- Typically, one queen is placed in each column.

3. Actions (Operators)

At each step, the agent can:

- Place a queen in any safe row of the next column.
- Move to the next column after placing a queen.
- Backtrack if no safe position is available.

4. Goal Test

The goal is achieved when:

- All 8 queens are placed, and
- No two queens share the same row, column, or diagonal.

5. Path Cost

- Each queen placement has a cost of 1.
- Total cost to reach a solution = 8 placements.

	C1	C2	C3	C4	C5	C6	C7	C8
R1	Q	.	.	.	.	.	.	.
R2	.	.	.	.	.	.	Q	.
R3	.	.	.	.	Q	.	.	.
R4	.	.	.	.	.	.	.	Q
R5	.	Q	.	.	.	.	.	.
R6	.	.	.	Q	.	.	.	.
R7	.	.	.	.	.	Q	.	.
R8	.	.	Q	.	.	.	.	.

Performance Measure

A solution is considered successful if:

- All **8 queens** are placed.
- No queens attack each other.
- The solution is found with minimum search time and memory.

4.OLA Cab Booking – Problem Solving Agent

1. Type of Problem-Solving Agent

The Goal-Based Problem-Solving Agent is the most suitable.

Reason:

- The customer's goal is to travel from the source location to the destination.
- The agent searches through different cab options (Mini, Micro, Sedan, Shared, Prime, etc.).
- It selects the cab that best satisfies the customer's requirements such as comfort, fare, and availability.

2. Problem Formulation

Component Description

Initial State Customer enters pickup location and destination.

State Space Available cab types (Mini, Micro, Sedan, Shared, Prime, etc.).

Actions Search cabs, compare fares, select cab, confirm booking.

Goal Test Cab successfully booked and customer reaches destination.

Path Cost Fare, travel time, waiting time, comfort level.

3. Pseudocode

Algorithm OLA_Cab_Booking

Start

Input Pickup_Location

Input Destination

Search available cabs

Display:

Mini

Micro

Sedan

Shared

Prime

Input Customer_Choice

If Selected_Cab is Available Then

Book Cab

Display Driver Details

Display Estimated Fare

Display Estimated Arrival Time

Display "Booking Confirmed"

Else

Display "Selected Cab Not Available"

Suggest another available cab

End If

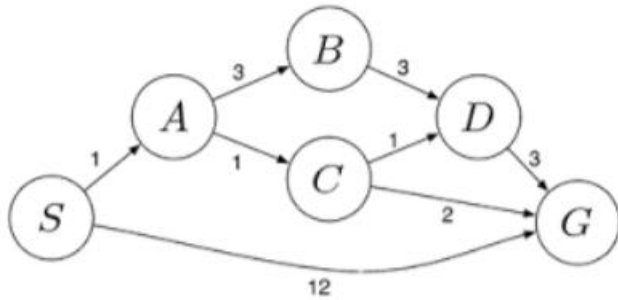
Stop

4. Working of the Agent

1. Customer enters **pickup** and **destination**.
2. Agent searches for available cabs.
3. Displays cab options:
 - Mini
 - Micro
 - Sedan
 - Shared
 - Prime
4. Customer selects a cab.
5. Agent checks availability.
6. If available:
 - Books the cab

- Shows driver details, fare, and arrival time
7. Otherwise:
- Suggests another available cab.

5. Uniform Cost Search (UCS) – Least Cost Path



Given weighted graph:

Edges and costs:

$$S \rightarrow A = 1$$

$$S \rightarrow G = 12$$

$$A \rightarrow B = 3$$

$$A \rightarrow C = 1$$

$$B \rightarrow D = 3$$

$$C \rightarrow D = 1$$

$$C \rightarrow G = 2$$

$$D \rightarrow G = 3$$

Step 1: Start at S

Priority Queue:

Node	Cost
S	0

Expand S

Add neighbors:

$$A (0+1) = 1$$

$$G (0+12) = 12$$

Queue:

Node	Cost
A	1
G	12

Step 2: Expand A (lowest cost)

Current cost = 1

Add neighbors:

$$B = 1+3 = 4$$

$$C = 1+1 = 2$$

Queue:

Node	Cost
------	------

C	2
---	---

B	4
---	---

G	12
---	----

Step 3: Expand C

Current cost = 2

Add neighbors:

$$D = 2+1 = 3$$

$$G = 2+2 = 4$$

Queue:

Node	Cost
------	------

D	3
---	---

B	4
---	---

G	4
---	---

G	12
---	----

Step 4: Expand D

Current cost = 3

Add:

$$G = 3+3 = 6$$

Queue:

Node	Cost
------	------

B	4
---	---

G	4
---	---

G	6
---	---

G	12
---	----

Step 5: Expand B

Current cost = 4

Add:

$D = 4 + 3 = 7$ (Ignored because D was already reached with lower cost 3)

Queue:

Node	Cost
G	4
G	6
G	12

Step 6: Expand G

Current cost = 4

Since G is the goal, UCS stops.

UCS Search Order

$S \rightarrow A \rightarrow C \rightarrow D \rightarrow B \rightarrow G$

Least-Cost Path

Trace back from G:

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost

$1 + 1 + 2 = 4$

Conclusion:

Using the Uniform Cost Search (UCS) algorithm, the least-cost delivery route from the starting warehouse S to the destination warehouse G is:

$S \rightarrow A \rightarrow C \rightarrow G$

with a minimum total travel cost of 4.